EE 5239 Guidelines for Course Project and Proposals

The course project is an important part of this course. It is designed to help you understand the materials you learnt throughout the course.

The project will be graded by the quality of the report. In the report you should clearly state the following: background of the problem; why the problem in general is interesting (why you picked the problem); which are the specific aspects that you want to model in your problem; what you have done to set up the model; which source of data you have used and why; what problems you have encountered during the modeling process and what are your strategies to deal with them; what's your solution method; How can you interpret your solution? Have you tried to model the problem differently and compare the solutions? Or if you set up your model differently will you arrive at the same solutions?

For the project proposal, it does not need to be long (e.g., 1 page). Below is a list of things that you might consider covering in your proposal. Also when you are writing your proposal and thinking about project ideas, keep in mind the these as guidelines for judging your final work. Also it is encouraged if the topic is related to your research.

1) Who are you working with (2-person max per group)
2) The problem you plan to work on and why
3) What are the aspects that are intrinsic to the problem that must be included in the modeling
4) What type of data do you expect that you will need, and where to find them
5) What mathematical models you may need in order to set up the problem
6) What programming language do you plan to use to solve the problem
7) What do you expect your solutions will look like (rough idea, qualitatively), what can you do in terms of modeling to improve your solution (finer model? better data? etc).

Some general suggestions on the project:

1) Implement some optimization algorithm from a paper, and show that you can reproduce the results in the paper.
2) Solve a real problem (possibly carved out of a research project you currently have).
3) Develop an algorithm and perform analysis.
4) Find a problem in Kaggle.com (a machine learning/data science problem/competition repository), or other similar open problems, and solve it.